



Joint ventures and technology adoption: A Chinese industrial policy that backfired

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ABSTRACT

To spur technology transfer, emerging market policymakers often require foreign firms to form joint ventures (JVs) with domestic firms. Through knowledge spillovers, JVs may reduce technology acquisition costs for domestic firms. Yet domestic firm rents from JVs could discourage innovation through a cannibalization effect. Which force dominates is an empirical question. I address it with novel data on China's auto sector. In response to fuel economy standards requiring firms to upgrade technology or sacrifice quality, firms with JVs reduced quality and price relative to their counterparts. Consistent with cannibalization, firms with JVs drive the negative effect.

China's policy of requiring all foreign car makers to form local joint ventures is "like opium" for Chinese firms and is failing to foster world-class indigenous automakers, a former minister was quoted as saying.
– Reuters (2012) quoting He Guangyan, Former Minister of Machinery & Industry.

1. Introduction

Important technologies have historically originated in developed countries and trickled down to lagging countries (Comin and Hobijn, 2004). To hasten this process, many emerging market governments encourage technology transfer from advanced foreign firms to backward local firms. I explore how an industrial policy designed to induce technology transfer perversely disincentivized technology acquisition. To my knowledge, this paper is the first quasi-experimental evaluation of an industrial policy's effect on firm technology acquisition in an emerging market.

Developing countries often require foreign entrants to form joint

ventures (JVs) with domestic firms. JVs are supposed to increase domestic partners' access to foreign R&D and manufacturing processes, reducing the cost of technology acquisition. JVs have a second important feature: the domestic partner receives a share of foreign brand profits. I show how these two features affect the domestic partner's innovation incentives in a stylized model. First, in an adaptation of the Gilbert and Newbery (1982) efficiency effect, the JV reduces the cost of technology acquisition. Pushing in the opposite direction is a cannibalization channel, similar in spirit to the Arrow (1962) replacement effect, which deters monopolists from innovating. The threat of cannibalizing rents from foreign partner sales discourages the domestic firm from investing in substitutes to its partner's products. This can be interpreted as an economic mechanism explaining why domestic firms with JVs might have lower absorptive capacities than their counterparts without JVs. These countervailing forces exist independently of the foreign partner's technology transfer behavior.

How JVs impact innovation is relevant to policy in many countries; for example, Brazil, Mexico, India, Nigeria, and Malaysia have employed JV mandates.² As a case study, I focus on China's automotive

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² See UNCTAD (2003) and Blomström et al. (2000).

industrial policy, which has called for globally competitive, high-quality Chinese firms since the late 1970s.³ The policy's cornerstone is a mandate that foreign entrants produce via JVs with domestic firms. High tariffs precluded large-scale imports, so foreign brands must establish JVs in order to access China's market.

The following example illustrates China's JV structure. Ford manufactures in China through two JVs. The larger is with Chang'an Automobile group. The Chang'an Ford JV plants produce only Ford vehicles. Chang'an helped finance the JV plants and receives fifty percent of their profits. Chang'an produces its own brands at other plants without Ford involvement. Foreign brands like Ford, Volkswagen, and Toyota have consistently dominated China's market in quality, price, and market share. Chinese firm exports to developed markets have been negligible. The failure of China's auto industrial policy to produce brands that can compete even domestically is a puzzle that goes beyond the inefficiencies associated with state ownership. In this paper, I evaluate whether the JV mandate achieved the explicit government objective of technology acquisition and upgrading among domestic firms.

China's sudden and stringent 2009 fuel economy standards provide plausibly exogenous variation in the fixed cost of technology upgrading.⁴ The standards compelled automakers to incur technology upgrading costs if they wished to produce high quality vehicles, which tend to be heavy and powerful. That is, an automaker facing fuel economy standards can either augment fuel efficiency technologies or reduce quality. The fuel economy policy imposed a fixed cost disadvantage on domestic firms because foreign firms (e.g. Ford), which already faced fuel economy standards elsewhere, incurred only the variable cost of including their efficiency technologies in local production.

I use a difference-in-differences design with novel, reliable, comprehensive model-level sales and characteristics data for the Chinese auto market between 1999 and 2013. To proxy for quality, I use four product attributes: torque (force the engine can exert, or turning power), price, weight, and the power-to-weight ratio (closely related to acceleration). These are well-established as measures of quality in the automotive sector. Product attributes offer an alternative to the conventional measures used in the literature, such as accounting-based productivity functions, R&D investment, and patents. These are more opaque measures, only distantly connected to the firm's actual products. More importantly, they are of little use in China, in part due to different patenting and data collection cultures.

While foreign firms continued on an upward trajectory, China's standards led domestic firms to reduce quality and price, without gaining market share. Variation is within firm, and foreign firms are treated as the “control” in most specifications. I assume foreign firm technology transfer behavior did not change immediately around the policy. Relative to foreign firms, the policy reduced domestic model price by 15 percent, torque by 11 percent, weight by 5 percent, and power-to-weight by 6.5 percent. I confirm the main result in a triple-differences design exploiting the standards' staged implementation in 2008 and 2009 for new and continuing models. Robustness tests, including placebo, alternative time spans around the policy, and varying fixed effects provide further confirmation. I also demonstrate pre-policy parallel trends across firm types.

The negative effect is strongest for firms with JVs. It is present but smaller for state-owned enterprises (SOEs) without JVs, and insignificant for private firms. Although few firms are SOEs without JVs or private with JVs, I establish significance across effects in specifications that interact the policy with firm status. The policy's effect is 16–18 percent larger among firms with JVs than among SOEs. This analysis

assumes that changes in model attributes across domestic firms with and without JVs would have been the same after relative to before the change, had the policy not occurred.

If a cannibalization threat disincentivizes innovation, the effect should be larger among domestic firms that compete more intensively with their partner. This concept is similar to the diversion ratio in merger analysis of differentiated products, as in [Shapiro \(1996\)](#). Indeed, I show that the policy had much stronger negative effects on domestic firms that operated in similar price segments or vehicle classes as their foreign JV partners prior to the policy. In sum, the negative effect of increasing own quality on the share of JV profits appears to outweigh any advantage from knowledge spillovers.

The JV mandate and the fuel economy policy were successful in two senses: foreign firms brought new technology to China, and fuel efficiency improved. However, *both* policies explicitly aimed to increase technology upgrading among domestic firms. I find that both had precisely the opposite effect. This contrasts with the government's intentions. My findings are consistent with the literature documenting that (a) private firms are more productive than SOEs in China; and (b) JVs are negatively correlated with technology diffusion.⁵ JVs can lead domestic firms down the manufacturing quality ladder, helping to reconcile FDI's positive role in the endogenous growth literature with mixed empirical findings at the country level.⁶

This paper argues that the JV policy “failed” *only* from the perspective of the government's explicit goals that the JV mandate encourage domestic firm innovation. I do not address the JV mandate's broader welfare effects, including on employment, brand variety, and government revenue. I also cannot assess whether the post-fuel economy policy decision to go down-market was profit maximizing. Further, my analysis addresses short term responses to a technology cost shock. In the longer term, domestic Chinese firms may reach the global technology frontier; my results suggest these will likely be private firms without JVs. Finally, my empirical results may not generalize to voluntary JVs or those in which partners have similar technical capacity. However, the cannibalization channel that I propose may exist in JVs more broadly, and my results do indicate the difficulty of contracting knowledge spillovers.

Despite these limitations, this paper contributes to our understanding of government's mediating role in technology diffusion, which is central to economic development ([Young, 1991](#); [Lucas, 1993](#)). A story in which JVs lead domestic firms *down* the manufacturing quality ladder helps to reconcile FDI's positive role in the endogenous growth literature with mixed empirical findings at the country level, where industrial policy regulates FDI.⁷ More broadly, my results speak to a debate about post-World War II growth. New growth theory advocates trade and investment openness to close technology gaps ([Coe and Helpman, 1995](#); [Baldwin, 1969](#)). Conversely, new institutional economists attribute the success of East Asian “Tigers” to government direction ([Rodrik et al., 2004](#); [Amsden \(1991\)](#)). In my setting, the most innovative firms are the least touched by industrial policy.

I also contribute to the literature about JVs, which has found both positive and negative effects on participating firm outcomes ([Lyles and Salk, 1996](#); [Inkpen and Crossan, 1995](#)). The literature on FDI in China has not addressed domestic partner learning ([Lin et al., 2009](#); [Nam, 2011](#); [Du et al., 2011](#)). In developed country context, research has focused on the potential for joint ventures, particularly those focused on

³ On (a), see [Lin et al. \(1998\)](#), [Allen et al. \(2005\)](#), [Khandelwal et al. \(2013\)](#), [Chen et al. \(2015\)](#), and [Fang et al. \(2015\)](#). On (b), see [Ramachandran \(1993\)](#), and [Moran \(2002\)](#). However, other studies find evidence of positive spillovers from JVs, like [Dimelis and Louri \(2002\)](#) and [Javorcik \(2004\)](#).

⁶ See [Bloom et al. \(2016\)](#) and [Aitken and Harrison \(1999\)](#). Key theoretical work includes [Bardhan \(1971\)](#), [Romer \(1993\)](#), and [Melitz \(2005\)](#). Related to this paper is [Müller and Schnitzer's \(2006\)](#) theoretical model of technology transfer in international JVs.

⁷ On industrial policy broadly, see [Grossman and Helpman \(1994\)](#), [Nunn and Trefler \(2010\)](#), and [Arnold and Javorcik \(2009\)](#). For the mixed results on FDI, [Hale and Long \(2011\)](#) for a review.

³ See, for example, State Council.

⁴ China imposed fuel economy standards in phases from 2005 to 2009, but binding standards came into force in 2009 (see Section 5).

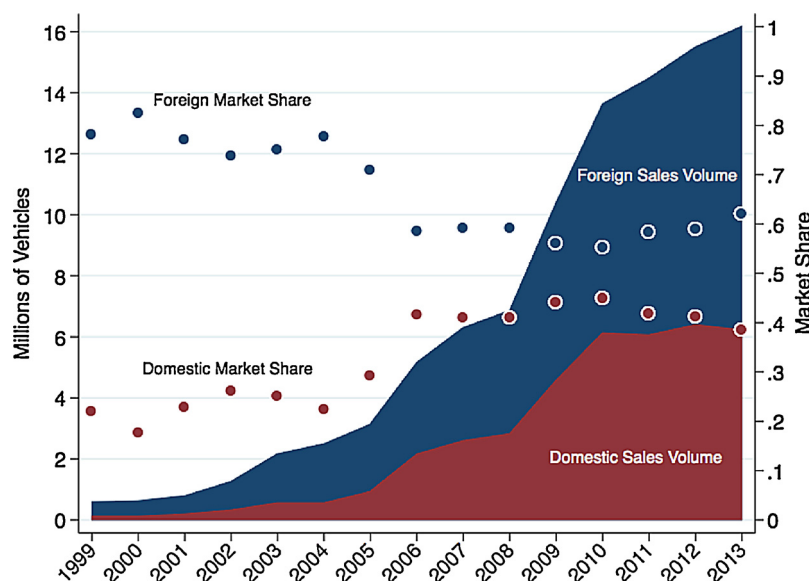


Fig. 1. Sales volume and market share by firm type. Note: This figure shows foreign and domestic brand Chinese sales volume (number of new vehicles sold in a given year) on the left axis, and market share on the right axis.

innovation investment, to achieve efficiencies (e.g. Vonortas, 1997; Gugler and Siebert, 2007). It has also explored the ability for firms to collude via R&D-focused joint ventures (Kamien et al., 1992; Duso et al., 2014).

While I do not estimate a counterfactual to the JV mandate, China could clearly have pursued alternatives. One option was to liberalize foreign firm entry and imports. The electronics sector, where China placed fewer constraints on FDI and permitted freer competition, illustrates the potential for rapid growth and dynamic indigenous firms, such as Xiaomi or Lenovo. A second path is Japan and Korea's combination of infant industry protection, foreign technology licensing, and reverse engineering. This path is more difficult under modern trade law, but China arguably pursued it successfully in internet services, where protectionism combined with freewheeling competition among privately owned startups produced Tencent, Baidu, and Alibaba.

Despite a rich theoretical literature, it has been challenging to evaluate industrial policies that target technology upgrading. I depart from much of the past literature by using technical quality. Firm-level panel data are also relatively rare in the literature on technological capacity and innovation, which has relied on aggregates, case studies or cross-sectional survey data, particularly for the developing world (Fagerberg et al., 2010). This paper is related to the literatures on inefficiencies in China's industrial structure (Hsieh and Klenow, 2009; Khandelwal et al., 2013) and evaluations of subsidy programs (Rotemberg, 2015; Howell, 2017).⁸

The paper proceeds as follows. I provide historical context about the Chinese auto sector in Section 2. Section 3 presents a simple model of innovation incentives in a JV. I describe the data in Section 4, and the empirical strategy data in Section 5. The results are in Section 6, and robustness tests in Section 7.

2. Industry context

This section first provides historical background on China's joint venture (JV) policy and how it has related to the development of its auto industry (Section 2.1). Then it explains the fuel economy policy (Section 2.2).

2.1. The JV policy

Since the late 1970s, China has vigorously deployed industrial policy in the service of building a globally competitive, high quality indigenous auto sector. In 1986, the central government designated the automotive sector a "Pillar Industry," and it has subsequently described automobile production as key to China's development.⁹ Beijing permitted FDI in automobile manufacture only via partnerships with domestic firms, which were supposed to evolve into globally competitive multinationals (State Council, 2006). High tariffs historically precluded imports (see Appendix Fig. A1 and its note), so a JV was the door to China's market for a foreign firm.

The JV is a stand-alone enterprise producing only foreign brand cars, and the foreign firm designs, controls, and operates the plant.¹⁰ The foreign firm owns no more than 50 percent, and usually also retains 50 percent of profits. Beyond the even profit split, there is no public information about contractual relationships. The government initially handpicked domestic partners, but after WTO accession in 2001 retreated to approving JVs (Richet and Ruet, 2008). Following WTO accession, the government removed barriers to entry for private firms. As in other sectors, SOEs were corporatized, largely separated from direct government control, and often partially listed on stock exchanges (Andrews-Speed, 2012). Today, domestic Chinese auto manufacturers exist along two axes: whether or not they are majority state-owned, and whether or not they have JVs with foreign firms. The literature on China's economy has focused on SOE efficiency relative to private firms.¹¹ However, in some high-tech sectors, SOEs have become globally competitive, dominating the domestic market and achieving meaningful exports.¹² Hsieh and Song (2015) show that in the 2000s SOEs had faster TFP growth and higher labor productivity than private firms.

⁹ The 1986 7th Five-Year Plan instructed policymakers to consider the "automotive industry as an important pillar industry, and it should follow the principles of 'high starting point, mass production, and specialization' to establish backbone enterprises as leaders" (Chu, 2011). The most recent automotive sector plan states that "Development of the automobile industry, including transformational upgrading, is an urgent task and is important for new economic growth and international competitive advantage" (State Council, 2012).

¹⁰ This applies to the period I study, from 1999 to 2013. In recent years, some JVs have produced new, joint brands.

¹¹ See Khandelwal et al. (2013), Bajona and Chu (2010), Jefferson et al. (2003), and Lin et al. (1998).

¹² Examples are wind turbine company Goldwind, shipbuilding company China State Shipbuilding Corporation, and steel manufacturer Baosteel.

⁸ This paper is more distantly connected to the literature on attribute-based regulation (e.g. Aldy and Houde (2015); Ito and Saltee, 2015).

Table 1
Model-level summary statistics by firm type.

	Price (RMB) ^a	Sales volume	Sales-wtd price (¥)	Sales-wtd price (\$)	Torque ^b	Norm. torque ^c	HP/ weight ^d	Weight (kg)	Height (mm)	Length (mm)
<i>Panel 1: All firms</i>										
Mean	136,089	31,209	122,340	16,834	170	4.33	1.90	1332	1546	4422
Median	103,900	13,882	109,169	14,870	156	3.80	1.91	1304	1483	4500
Std dev	103,755	55,186	81,684	11,694	59.7	2.19	0.25	298	156	416
N	3128	3177	3194	3087	2726	2715	2624	2643	2723	2724
<i>Panel 2: Foreign (non-Chinese) firms pre-policy</i>										
Mean	168,673	24,574	151,268	20,191	179	4.62	1.94	1349	1512	4456
Median	131,800	11,867	133,618	17,655	170	4.00	1.94	1304	1472	4520
Std dev	112,308	33,417	85,141	11,400	57.6	2.26	0.19	283	107	395
N	437	438	438	436	405	405	401	405	410	410
<i>Panel 3: Foreign (non-Chinese) firms post-policy</i>										
Mean	171,892	33,560	153,595	22,669	190	4.81	1.99	1382	1520	4492
Median	134,800	17,302	137,528	18,510	177.5	4.08	1.98	1386	1484	4540
Std dev	114,900	43,482	78,291	12,097	62.4	2.69	0.29	282	101	363
N	639	644	644	642	610	610	598	598	610	610
<i>Panel 4: Domestic (Chinese) firms pre-policy</i>										
Mean	83,763	25,845	82,704	9860	146	4.04	1.78	1285	1606	4338
Median	74,300	7508	77,669	8385	143	3.42	1.83	1200	1521	4434
Std dev	46,675	56,317	40,805	5837	48.7	2.19	0.25	322	186	482
N	280	290	287	253	233	231	221	221	225	225
<i>Panel 5: Domestic (Chinese) firms post-policy</i>										
Mean	73,231	37,315	66,496	9645	139	3.62	1.74	1261	1597	4291
Median	66,350	13,289	64,932	9593	136	3.20	1.77	1200	1495	4375
Std dev	32,085	96,082	27,022	4425	41.4	1.77	0.20	267	225	438
N	436	448	446	426	398	397	375	375	385	386

Note: This table shows summary statistics at the model-year level.

^a Nominal RMB.

^b Maximum torque, in nanometers.

^c Torque specified at a particular speed, or rotations per minute (rpm). More power at lower speed is better, so lower RPM is better.

^d Horsepower in kw divided by weight/100.

In negotiations to establish initial JVs, foreign firms benefited from information asymmetry about auto manufacturing. They sought to bound potential technology transfer by initially producing only out-dated models in China, by keeping high end part production overseas, and through other, less observable ways (Oliver et al., 2009). In response, a 1994 policy directive required JVs to have “the capacity for manufacturing products which attain the international technological levels of the 1990s” as well as an R&D center (Walsh, 1999).¹³ The 1994 directive and similar policies were unenforceable. Instead, competition compelled foreign firms to produce the latest models in China by the mid-2000s.

Incomplete contracting and moral hazard bedeviled implementation of the JV arrangements (Thun, 2004). For example, most GM-branded models initially chosen for China were Daewoo or Opel designs, distancing GM's China operation from Detroit's state-of-the-art. Though GM marketed itself as a purveyor of useful technology, its China research center was largely used to tweak existing models for the Chinese market (Tang, 2012). In a paper related to this one, Nam (2011) uses detailed case studies to investigate international JVs in China's auto sector. Nam concludes that

“The IJV arrangement in itself provides local firms with only “passive” and “incomplete” learning opportunities because foreign firms, which have superior technological capabilities, can effectively control various aspects of the main access channel to their strategic assets (knowledge and skills, in particular)...In most cases, MNCs [foreign firms] have provided their IJVs with the explicit “outcomes” of their technological capabilities, not the technological capabilities themselves. The IJV arrangement has discouraged local firms from making efforts to internalize the transferred knowledge for their own good.

¹³ WTO terms forbid market access-technology transfer *quid pro quo*, but the stated technology transfer requirements remain in place.

Foreign firm behavior is not the focus of this paper, but it appears consistent with Branstetter and Saggi (2011), who theorize that stronger intellectual property rights reduce imitation risk, increasing FDI and innovation incentives.

Beijing has consistently called for “self-reliant Chinese car manufacturers who rank among the 500 largest global firms” (NDRC, 2004). Yet foreign brands dominate China's passenger vehicle market by common quality proxies, such as sales, price, torque, power-to-weight, weight, height, and length (see Fig. 1, Appendix Fig. A2, and Table 1). Surprisingly, foreign dominance persists across the four major vehicle classes (compacts, minivans, SUVs, and sedans). Appendix Table A1 contains *t*-tests comparing foreign and domestic models within each class. It shows that, for example, the sales volume of foreign compact model-years has on average been 20,207 for foreign brands, and 18,411 for domestic brands, despite the fact that the average foreign compact model is 40 percent more expensive, on a sales-weighted basis, than the average domestic compact model. Foreign compact models have significantly higher torque, weight, and length. These differences only grow more extreme for the other three classes.

Though China has been the world's largest passenger vehicle market since 2010, the economies of scale that characterize the global auto industry have thus far eluded Chinese firms. There is ample anecdotal evidence from the popular press that JVs failed to achieve technology transfer.¹⁴ Dunne (2012) concludes that:

“Chinese auto regulators find themselves in a tight spot: their 30-

¹⁴ Ying (2012) quotes Liao Xionghui, the Vice President of private Chinese automaker Lifan, as saying: “We have been trying to exchange market access for technology, but we have barely gotten hold of any key technologies in the past 30 years.” Ho (2015) writes that: “Requiring foreign carmakers to form ventures at least 50 percent-owned by a Chinese partner had an explicit goal to create three or four internationally competitive homegrown auto giants by 2010. Instead, the policy has drawn criticism for shielding state-owned carmakers from competition and robbing them of the incentive to build their own brands.” See also Gallagher (2006), Holmes et al. (2015), Economist (2013), and Sanford (2013).

year quest to build an industry dominated by Chinese car brands has backfired. The problem: joint ventures with foreign carmakers that have proven just a tad too comfortable.”

2.2. Fuel economy policy

My empirical strategy requires a shock to the costs of improving quality. I use the China's fuel economy policy, which compelled automakers to incur technology upgrading costs if they wanted to produce high quality vehicles, which tend to be heavy and powerful. In 2004, China's National Development and Reform Commission announced that China would adopt fuel economy standards, with two aims: (1) to decrease oil consumption for energy security purposes; and (2) to increase technology transfer by forcing foreign firms to bring more up-to-date technology to China (Wagner et al., 2009; UNEP, 2010). The standards were anticipated by automakers, who had time between 2004 and final implementation in 2008–2009 to tweak existing assembly processes to meet the standards and plan new models with the standards as a constraint.

There is a basic tradeoff between fuel economy, weight, and power. Knittel (2011) shows that among U.S. auto manufacturers, decreasing weight in passenger cars by 10 percent is associated with a 4.2 percent increase in fuel economy.¹⁵ Automakers can meet new standards without new technology by building lighter, less powerful cars. Margins for adjustment that reduce quality include replacing parts with flimsier, lighter parts and downsizing the engine. Alternatively, automakers can maintain quality and meet the standards by acquiring fuel efficiency technologies, including discrete engine parts like catalytic converters and whole-vehicle design improvements in the power-train, aerodynamics and rolling resistance.¹⁶ Heavier and more powerful vehicles generally have higher profit margins than other segments (IMF, 2006).

To maintain existing models' quality, the fuel economy standards compelled domestic automakers to acquire fuel efficiency technologies and integrate them into the model design. Conversely, foreign firms – which had faced stringent fuel economy standards in Japan and Europe for decades – merely incurred the variable cost of inserting technologies developed for other markets into their China production.

China implemented Phase 1 fuel economy standards in 2005 for new models and 2006 for continuing models. A number of studies conclude that these initial standards were not binding.¹⁷ Phase 2 came into effect in 2008 for new models, and 2009 for continued models. Phase 2 was more stringent than contemporaneous U.S. standards, but less stringent than Japanese or European standards (Appendix Table A2 lists the standards by weight class, and Fig. 2 compares standards across countries).¹⁸ My interviews in 2013 at the government-affiliated China Automotive Research and Technology Center (CATARC) in Tianjin, which was partially responsible for developing fuel economy standards and testing vehicles, confirmed that meaningful enforcement of the standards and consistent fuel economy testing began in 2008–2009.¹⁹ I use 2009 as the policy implementation year.

The Chinese standards are designed to be stricter for heavier vehicle classes (An et al., 2011). Before the standards, automakers selling vehicles in China did not have to report fuel economy. Further, government inspection and enforcement were lax prior to the Phase 2

implementation. Since there is little data on fuel economy before the standards, it is impossible to compare fuel economy before and after the standards. Fig. 3 shows that as of 2010, the vast majority of new models met the standards.²⁰

3. Empirical approach

This section first uses a simple model to explain the logic and incentives behind the effect of a joint venture on innovation outcomes, and the role of a technology cost shock (Section 3.1). This informs the specific reduced form empirical strategy, which is described in Section 3.2.

3.1. Incentives to innovate in joint ventures

This section proposes two possible opposing mechanisms through which JVs may affect innovation and product quality. The government intended for JVs to reduce the cost of acquiring technologies for domestic Chinese firms. If the government's assumption that JVs would lower the cost of acquiring foreign technologies was correct, then we should observe firms with JVs not reducing car quality when faced with an increase in technology requirements (the fuel economy standard). Below, I will term this channel the “efficiency effect”. Alternatively, the JV industry structure may attenuate innovation incentives if producing substitutes to foreign partner models cannibalizes the domestic firm's share of JV profits. This will be termed the “Arrow replacement effect.” Subsequent sections will assess whether the data are more consistent with one channel or the other.

Consider the following stylized profit functions for domestic firms:

$$\text{Firm without JV: } \pi_j = \sum_{i \in j} q_i(\mathbf{p}, \phi_i)(p_i - C_{i,\text{No JV}}) \quad (1)$$

$$\text{Firm with JV: } \pi_j = s\pi_{jV}^{\text{foreign}} + \sum_{i \in j} q_i(\mathbf{p}, \phi_i)(p_i - C_{i,\text{JV}}), \quad (2)$$

where j denotes firm, i denotes model, ϕ technology quality, and s the domestic firm's profit share from foreign model sales. I assume a competitive market, and that $\frac{\partial p_i}{\partial \phi_i} > 0$, $\frac{\partial q_i(\mathbf{p}, \phi_i)}{\partial \phi_i} > 0$ and $\frac{\partial q_i(\mathbf{p}, \phi_i)}{\partial p_i} < 0$. The firm's cost function ($C_i = F(\cdot, \phi_i)$) is also increasing in quality ($\frac{\partial C_i}{\partial \phi_i} > 0$). The first order conditions in quality are:

$$\begin{aligned} \text{Firm without JV: } \frac{\partial \pi_j}{\partial \phi_i} &= q_i(\mathbf{p}, \phi_i) \left[\frac{\partial p_i}{\partial \phi_i} - \frac{\partial C_{i,\text{No JV}}}{\partial \phi_i} \right] \\ &+ \frac{\partial q_i(\mathbf{p}, \phi_i)}{\partial \phi_i} (p_i - C_{i,\text{No JV}}) \\ &+ \sum_{k \neq i \in j} \left[\frac{\partial q_k(\mathbf{p}, \phi_k)}{\partial \phi_i} (p_k - C_{k,\text{No JV}}) \right]; \end{aligned} \quad (3)$$

$$\begin{aligned} \text{Firm with JV: } \frac{\partial \pi_j}{\partial \phi_i} &= s \frac{\partial \pi_{jV}^{\text{foreign}}}{\partial \phi_i} + q_i(\mathbf{p}, \phi_i) \left[\frac{\partial p_i}{\partial \phi_i} - \frac{\partial C_{i,\text{JV}}}{\partial \phi_i} \right] \\ &+ \frac{\partial q_i(\mathbf{p}, \phi_i)}{\partial \phi_i} (p_i - C_{i,\text{JV}}) \\ &+ \sum_{k \neq i \in j} \left[\frac{\partial q_k(\mathbf{p}, \phi_k)}{\partial \phi_i} (p_k - C_{k,\text{JV}}) \right]. \end{aligned} \quad (4)$$

The foreign firm's profit decreases in a competitor's quality ($\frac{\partial \pi_{jV}^{\text{foreign}}}{\partial \phi_i} < 0$). Thus the domestic firm's investment in own quality reduces its marginal profit from the JV.²¹ This is a version of the Arrow replacement effect (ARE): when a competitive firm and a monopolist have the same profits from an innovation, the monopolist has a lower

¹⁵ Some fuel efficiency technologies – particularly in the engine – may be outsourced to suppliers. Component suppliers are an important part of the automotive industry. However, they are an independent sector and beyond the scope of this paper.

¹⁶ Other specific technologies include reducing transmission losses, direct fuel injection, variable valve timing, turbochargers, superchargers.

¹⁷ See Wagner et al. (2009), Oliver et al. (2009), and An et al. (2011).

¹⁸ The Phase 2 standards are roughly equivalent to Euro IV. China uses the New European Driving Cycle (NEDC) testing method, rather than the CAFE method used in the U.S.

¹⁹ I met with Shi Jian and Liu Bin in the CATARC Auto Industry Policy Research Division.

²⁰ A Phase 3 program is currently underway that adds corporate average fuel economy targets to the weight-based system. According to the 2012 Energy-Saving and New Energy Vehicle Industrialization Plan, the goal is to achieve a fleet average of 6.9 L/100 km by 2015, and 5.0 L/100 km by 2020.

²¹ All firms have the same variable cost of producing more fuel efficient vehicles.

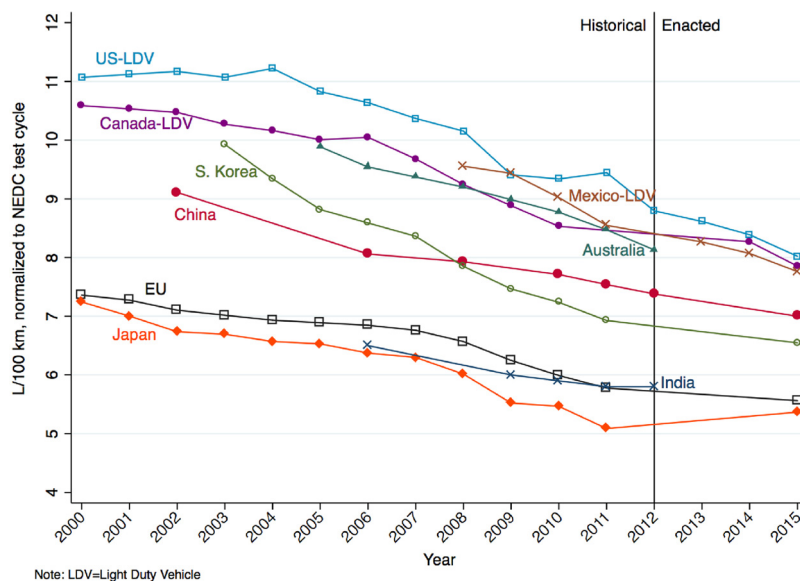


Fig. 2. Fuel economy standards by country, 2000–2015. *Note:* This figure shows historical and enacted fuel economy standards by country, in liters of gasoline per kilometer. Data from ICET (2013).

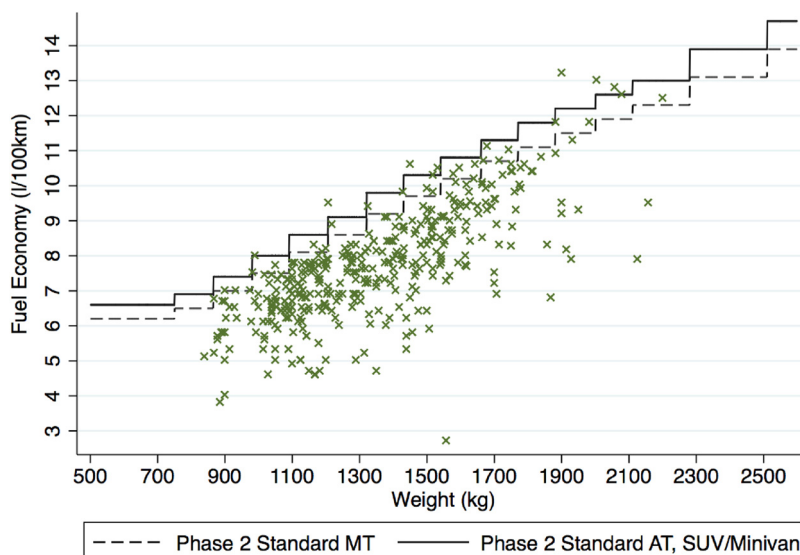


Fig. 3. Model fuel economy and weight, with Phase 2 standards, 2010. *Note:* This figure shows China's 2009 Phase 2 fuel economy standards. Dotted line is for manual transmission, line is for automatic and all SUVs/minivans.

incentive to invest in R&D because he earns profits on the existing technology that would be cannibalized by sales of the new technology (Arrow, 1962). Here the “monopolist” is the domestic firm with a JV, and the “competitive entrant” is the domestic firm without a JV. Suppose the fixed cost of achieving some technology quality level ϕ_i is $F_j(\phi_i)$, so that $C_i = \mathcal{F}(\cdot, F_j(\phi_i))$.²² The ARE stems from $\frac{\partial}{\partial \pi_{JV}^{\text{foreign}}} \left(\frac{\partial \pi_j}{\partial F_j} \right) < 0$ (see Fudenberg and Tirole, 2013). The more profit the domestic firm gets from its JV, the lower the incentive to spend F_j to acquire the new technology.²³

Countering the ARE is what Gilbert and Newbery (1982) call the

²² The fixed cost of technology acquisition is to some degree spread across models. I abstract from this here, though it applies if we assume equal spreading and equal number of models across firms.

²³ Note that the cannibalization effect also exists across models within each firm. When a firm expands its automotive technology frontier it generally takes market share from incumbents in a higher-value segment, and my focus here is on the JV effect on the within-firm choice to expand that frontier.

“efficiency effect.” They assume the monopolist is more efficient in making profits than a duopoly, so its preemptive payoff is larger than the entrant’s. The parallel assumption in the JV context is that domestic firms with JVs have a lower F_j because they have greater access to foreign firm technology. Therefore, $F_{j,JV} \leq F_{j,NoJV}$. Note that this assumption will push against my ultimate finding, which is that domestic firms with JVs are less able to innovate than those without JVs. This assumption is based on the government’s rationale for the policy, and is one that I impose on the data. Holding other aspects of the cost function fixed, $F_{j,JV} \leq F_{j,NoJV}$ implies that increasing quality is at least as costly for firms without JVs as for firms with JVs, or $\frac{\partial C_{i,JV}}{\partial \phi_i} \leq \frac{\partial C_{i,NoJV}}{\partial \phi_i}$. If the efficiency effect dominates, this will make the preemptive payoff higher, and the firm with a JV will invest more in innovation.

The fuel economy standards exogenously impose higher $F_j(\phi)$, or fixed cost for a given quality. To comply and maintain quality, domestic firms must invest this fixed cost, as well as bearing the variable cost, which I assume to be equal across firms. Foreign firms already possess the technology, because they had faced stringent fuel economy

standards in Japan and Europe for decades, so $F_{foreign} = 0$. The theory is ambiguous about whether firms with or without JVs have a greater incentive to invest in higher ϕ_i (pay the fixed cost F_j to acquire fuel efficiency technology). It depends on whether the negative effect on ϕ of access to the foreign firm's profits (ARE) outweighs the positive effect of a lower technology acquisition cost (efficiency effect). If the former cannibalization effect dominates, then:

$$\frac{\partial \phi_{i,JV}}{\partial F_{j,JV}} < \frac{\partial \phi_{i,NoJV}}{\partial F_{j,NoJV}} \text{ if } s \left[\frac{\partial \pi_{JV}^{foreign}}{\partial \phi_i} \right] < \frac{\partial C_{i,JV}}{\partial \phi_i} - \frac{\partial C_{i,NoJV}}{\partial \phi_i}. \quad (5)$$

The three terms on the left are negative. The two terms on the right are positive. The model also implies that among domestic firms with JVs, those that compete more intensively with their partners should be less incentivized to upgrade (they have a more negative $\frac{\partial \pi_{JV}^{foreign}}{\partial \phi_i}$).

If an automaker just meets the standards, its profit function contains a constraint generating a shadow cost of producing fuel inefficient vehicles. In his model of the U.S. auto sector, [Jacobsen \(2013\)](#) incorporates this shadow cost. Jacobsen and other structural models of fuel economy, like [Bento et al. \(2009\)](#), are much more complex than the present approach. Yet they treat technological change as exogenous to the competitive equilibrium. Firms respond to standards by changing prices, which shifts the vehicle mix to lower margin, more efficient models. However, multinational automakers have made large R&D investments in fuel efficiency ([Knittel, 2011](#)). Unlike much of the literature on fuel economy standards, I am only interested in their relationship to technology upgrading. They require technology upgrading investment to produce a given vehicle quality level. This fixed cost is zero before the standard, and zero for foreign firms. It is an opportunity to observe whether technology upgrading is more costly among firms with or without JVs (that is, whether $\frac{\partial \phi_{i,JV}}{\partial F_{j,JV}} < \frac{\partial \phi_{i,NoJV}}{\partial F_{j,NoJV}}$ or vice versa).

3.2. Empirical strategy

This section describes the empirical strategy that I use. Whether JVs cause domestic firms to innovate more or less depends on whether the negative effect of foreign partner profits outweighs the positive effect of lower technology acquisition cost. In a difference-in-differences (DD) design, I compare foreign and domestic firms' model characteristics before and after the 2009 fuel economy policy.

The standard DD design involves two groups, one of which is subject to a treatment in the second of two time periods. If the two groups are ex-ante similar and have similar time trends, then inclusion of controls for treatment and state should yield an estimated coefficient on the treated state that is the average difference between the treatment group and the control group. The fuel economy policy, which put domestic firms at a fixed cost disadvantage in technology upgrading, is the treatment. Foreign firms are the "control;" they already possessed fuel efficiency technology. However, since the policy applied to all firms, the estimated treatment effect is best interpreted as the difference in responses across firm types.

In 2004, China's National Development and Reform Commission announced that China would adopt fuel economy standards, with two aims: (1) to decrease oil consumption for energy security purposes; and (2) to increase technology transfer by forcing foreign firms to bring more up-to-date technology to China ([Wagner et al., 2009](#); [UNEP, 2010](#)). The standards were anticipated by automakers, who had time between 2004 and final implementation in 2008–2009 to tweak existing assembly processes to meet the standards and plan new models with the standards as a constraint. There is a basic tradeoff between fuel economy and vehicle quality (primarily weight and power). An in-depth discussion of this tradeoff and the standards is in the Appendix. To maintain existing models' quality, the fuel economy standards compelled domestic automakers to acquire fuel efficiency technologies and integrate them into the model design ($F_j(\phi_i) > 0$). Foreign firms

incurred only the variable cost of inserting technologies developed for other markets into their China production.

DD estimators pose two potential problems. First, the design fails if the policy is endogenous to the group studied. The fuel economy policy affected both foreign and domestic firms, and I have not found other policies or market structure changes in the period analyzed that affected only domestic firm production. Second, serial correlation in variables may cause downward bias in the standard errors, especially with a relatively long time series and DD implementation via time fixed effects. Pooling the data on either side of the treatment and clustering standard errors by group rather than time solves the problem, particularly when the number of groups is large.²⁴ I first estimate the following regression:

$$Y_{it} = \alpha + \beta(\text{Policy}_i \cdot \text{Domestic}_j) + \gamma_1 \text{Policy}_i [+ \gamma_2 \text{Domestic}_j] \\ [+ \mathbf{1}|\text{Firm/Model/Class} = j/i/c] + \varepsilon_{ijt}. \quad (6)$$

Each observation is a model-year, where i denotes the model (such as the BYD F6 or the Chevrolet Spark), j the firm (such as Chery or Honda), and t the year.

The outcome Y_{it} is log torque, log price, weight, or RCL quality. In specifications that exclude firm fixed effects, I include Domestic_j , which is 1 if the firm is Chinese (such as BYD or Chery), and 0 if it is foreign (such as Nissan or GM). Policy_i is 1 if the year is 2009 or later, and 0 otherwise. In the primary specification, I use three years of data on either side of the policy and firm fixed effects. In alternative specifications, I use model or class (compact, sedan, minivan, or SUV) fixed effects. The fairly short time period raises the possibility that I observe a transition period, and some firms (i.e. those with JVs) may require longer to transition. However, domestic firms with JVs have underperformed for decades, suggesting that my findings reflect a problem with their technology acquisition capabilities, not simply a longer transition period. I present evidence of pre-policy parallel trends and against selection into treatment in Section 7. Section 7 also contains a triple-difference specification exploiting the staged policy for new and continuing models, which addresses a number of the potential alternative stories. The coefficient of interest β is the effect of the policy on domestic firms relative to foreign firms.

I then interact the policy effect with JV and SOE status. A quadruple interaction estimates the effect of the policy on domestic firms that are SOEs with JVs, and triple interactions estimate the effect on private firms with JVs and SOEs without JVs.

$$Y_{it} = \alpha + \beta_1(\text{Policy}_i \cdot \text{Domestic}_j \cdot \text{Has JV} \cdot \text{SOE}_j) \\ + \beta_2(\text{Policy}_i \cdot \text{Domestic}_j \cdot \text{Has JV}_j) \\ + \beta_3(\text{Policy}_i \cdot \text{Domestic}_j \cdot \text{SOE}_j) + \beta_4(\text{Policy}_i \cdot \text{Domestic}_j) \\ + \beta_5(\text{Domestic}_j \cdot \text{Has JV} \cdot \text{SOE}_j) + \gamma_1 \text{Policy}_i [+ \gamma_2 \text{Has JV}_j] + \gamma_3 \text{SOE}_j \\ + [\mathbf{1}|\text{Firm/Class} = j/c] + \varepsilon_{ijt}. \quad (7)$$

The coefficients of interest are β_1 through β_4 , where β_4 gives the effect for private firms without JVs. I exclude the domestic individual effect because while no SOEs are foreign, all foreign firms have JVs. Thus with firm fixed effects $\gamma_2 \text{Has JV}_j$ is not identified. I also omit foreign firms to examine the effect of the policy when $\text{Domestic}_j = 1$. While this approach lacks the strength of the foreign firms as "control," it provides a clear test of the ARE.

If the cannibalization channel dominates, the policy's negative effect should be strongest among domestic firms that compete most directly with their JV partner. I test the effect of pre-policy competition using variation in market segment and vehicle class. First, since there are effectively no Chinese models competing in the luxury market, I limit the sample to down-market segments. I estimate Eq. (7) on models with

²⁴ See [Bertrand et al. \(2004\)](#). I cluster standard errors in 78 groups (firms).

below median price, torque, and weight. Second, I omit firms whose partners produce only high-end vehicles.²⁵ Third, I proxy for competition by splitting the sample according to whether a domestic firm and its foreign partner both produced above median levels of a certain vehicle class prior to the policy. I examine the three minority classes: SUVs, minivans, and compacts.²⁶ I estimate Eq. (7) on partner firms competing in the same class. An example of such a pair for SUVs is BAIC (“Beiqi”) Zhanqi and partner Hyundai. For minivans, an example is Jiangling Motors (JMC) Landwind and partner Ford. The cannibalization effect should be larger for such pairs.

The alternative to the DD approach in Eqs. (6) and (7) is a detailed structural model, as in Berry et al. (1995), Austin and Dinan (2005), and Allcott and Wozny (2014). Such a model could incorporate a lower fixed cost of improving fuel economy for firms with JVs, while estimated substitution patterns would predict more diversion from the foreign partner. Counterfactuals without the cannibalization channel could be estimated. However, such models assume profit maximization and Bertrand competition, which are not appropriate in the Chinese context. The natural experiment approach that I take is better suited to analyzing incentives for technology upgrading in a mixed economy.

4. Data and descriptive statistics

This paper is based on novel, national, model-level data of light-duty passenger vehicle sales in China between 1999 and 2013. These data are from police registration via the State Council Development Research Center, which conducts analysis for China's top-level State (i.e. not Party) governing apparatus.²⁷ Each observation is a new model-year, such as the 2010 Volkswagen Jetta. The data include the ultimate Original Equipment Manufacturer (OEM), brand, model name, vehicle class, engine displacement, and power train.²⁸

I acquired the following new model-year characteristics: price (MSRP), maximum torque (nm), peak power (kw), curb weight (kg), length (mm), height (mm), and fuel economy (l/100 km).²⁹ Unfortunately, fuel economy is unavailable before the fuel economy standards were implemented, so I do not use it when I compare models before and after the policy. Descriptive statistics at the model-year level are in Table 1.³⁰ Fig. 1 shows that sales increased from under 1 million units in 1999 to 16 million units in 2013. Variety increased as well; the number of models increased linearly from 23 in 1999 to 426 in 2012. Foreign market share declined from 80 percent in 1999 to 55 percent in 2009, and then increased somewhat to 61 percent in 2013 (these figures exclude imports).

²⁵ Specifically, I exclude partners of BMW and Mercedes-Benz (Beiqi, Beiqi Weiwang, Zhanqi, Brilliance).

²⁶ An alternative is to normalize the sales in a class by a firm's overall sales. I find somewhat similar results (not shown) using such a metric. However, note that in the first order condition (Eq. (4)), profit cannibalization depends on the change in foreign firm profits with respect to a quality change in the domestic firm's model. This in large part depends on the number of vehicles in that class (e.g. SUVs) that the foreign firm sells. The firm's fraction of sales that are SUVs is therefore a less direct measure of competition.

²⁷ Consumers (private and public) must register new vehicle purchases to the local police. I acquired these data in my capacity as a visiting scholar at the DRC (中国发展研究基金会), which was possible because of an invitation secured by Harvard Kennedy School Professor Anthony Saich from Lu Mai, the Secretary General of the DRC. The data itself was provided through the head researcher at DRC's Institute of Market Economy.

²⁸ OEM refers to the firms that design, assemble and brand vehicles such as Ford and Hyundai. There are four vehicle classes: compact, sedan, minivan, and SUV. Engine displacement is in liters. Powertrain is either internal combustion engine, natural gas, electric, or hybrid electric.

²⁹ This was done via web scraping. There is coverage for 82 percent of models (slightly more for foreign models (88 percent) than domestic (73 percent), and slightly better in later years). Models without characteristics have much lower sales; the mean sales volume is 13,629 for models lacking characteristics data compared with 25,824 for models with characteristics data.

³⁰ Versions of the same model with different engine sizes are not treated as different models.

I use four measures of quality. The first quality measure is price, which indicates quality in the auto market (Brucks et al., 2000). I use nominal local currency (RMB) prices to avoid issues around exchange rates (inflation is differenced out). Table 1 shows nominal RMB and dollar prices. The second measure is vehicle weight. In general, heavier cars have more amenities and are safer. One way to meet fuel economy standards without adding costs, such as through more advanced materials like aluminum, is to reduce weight. This will typically result in flimsier, smaller cars with fewer amenities. Vehicle torque, responsible for acceleration and power, is the third measure.³¹ Torque depends on the engine, transmission ratios, weight, and other aspects of vehicle integration. A car with more torque will have a better driving feel, and usually better engineering and design. I use maximum torque divided by the listed RPM.³² An alternative measure of acceleration is power relative to weight. Therefore, in the main specifications, I also show the results using power (in kilowatts) divided by weight, which I first normalize by 100 so that the units are not grossly different. Together, torque, weight and price provide an objective, publicly observable measure of vehicle quality (Santini and Anderson, 1993; EPA, 2015; Gramlich, 2010).

I restrict analysis to 3177 models with sales of at least 1000 units. Of these, I have price for 3128, torque for 2726, weight for 2643, and RCL quality for 2697. Appendix Table A3 shows correlations among the variables. By construction, torque and horsepower are closely connected: after normalizing for RPM, their correlation is 0.96. Sales volume is negatively correlated with price and quality. The correlation between weight and price is 0.67, and between torque and price is 0.51. I log torque and price because they exhibit significant positive skewness, while the distribution of weight is roughly normal and, importantly, symmetric. All three densities are in Appendix Fig. A3.

I use brands, such as Ford, Audi, BYD, and Roewe, as the primary unit of analysis. To avoid confusion, I term brands “firm,” but some are subsidiaries of one OEM. Brands are the unit of observation most relevant to quality. Design, engineering and final assembly generally take place at the brand level, especially in China, where some OEMs are JVs producing domestic and foreign brand vehicles, albeit at different plants. I show that my empirical results are robust to grouping at the OEM level. Appendix Fig. A2 shows that Chinese brands proliferated over time, while their sales-weighted dollar prices decreased. Despite lower prices, Chinese firms are not dominant in the low end of the market.

JVs are not randomly assigned, making it difficult to disentangle the effects of state ownership and JV status. One obvious channel through which JVs may be formed non-randomly is that firm innovation capacity may be correlated with the tendency to form a JV. While unobservable, such selection would be consistent with the paper's goal to evaluate the government's JV promotion. If only non-innovative firms select into JVs, the government should not be encouraging this behavior. Appendix Fig. A4 shows the number of active firms by type and year. My estimation relies on the small number of SOEs without JVs and private firms with JVs (six and three, respectively).³³ Appendix Table A4 replicates Table 1 among domestic firms, dividing them by JV and SOE status. Among SOEs there are 797 model-years, compared to 486 for private domestic firms. Among SOEs with JVs there are 456 model-years, compared to 82 for domestic private firms with JVs.

³¹ Torque is the amount of force the engine can apply in a rotational manner, measured in nanometers. Horsepower is torque multiplied by a given speed (rotations-per-minute, or RPM), and determines the top speed of the vehicle.

³² A model's advertised torque is the maximum achieved at a particular RPM. More power at lower speed is better, so lower RPM is indicative of higher quality. I also multiply by 100, because in native units RPM is about two orders of magnitude larger than torque.

³³ Unfortunately there are no consistent, reliable data on the share of a SOE that the government owns, so I cannot use this as a source of variation.

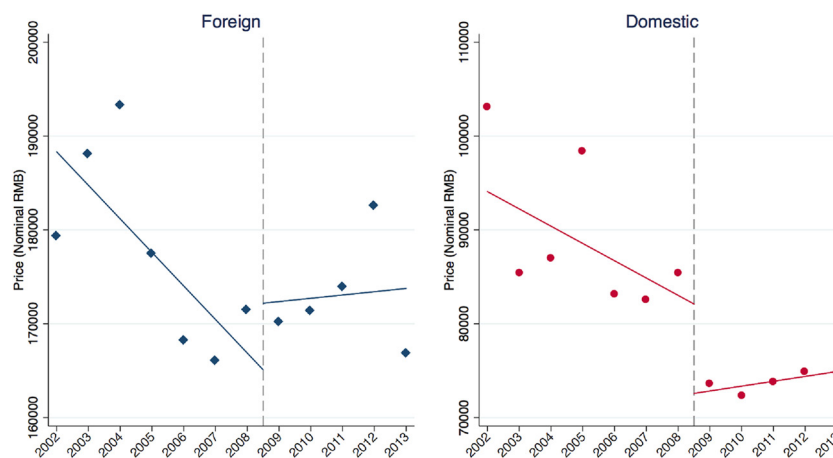


Fig. 4. Price by firm type, 2002–2013. Note: This figure shows binned scatterplots of price for foreign and domestic firms.

5. Results

This section describes the main results, beginning with visual evidence of the effect of the policy by firm type in Section 5.1. The primary regression results are in Section 5.2. I explore how the effects differ depending on how much a firm competed with its partner ex-ante in Section 5.3.

5.1. Visual evidence

Domestic firms – particularly those with JVs – responded to the 2009 fuel economy standard by manufacturing less powerful, cheaper, and lighter vehicles. Fig. 4 shows that domestic firms reduced prices in 2009, while foreign prices were unchanged (note the different axes).³⁴ Appendix Figs. A5 and A6 show log torque and log price in the three-year window around the policy. Among domestic firms, all three metrics fall in 2009, the year of the policy. Fig. 5 considers only domestic firms, and splits the firms by JV and SOE status. Panel A shows log torque, Panel B price, and Panel C power-to-weight. Firms with JVs appear responsible for the fall around the policy. For power-to-weight, the difference between the behavior of private firms with and without JVs is especially striking.³⁵

Empirically, exporting is strongly associated with firm productivity and competitiveness.³⁶ Chinese government policy explicitly encourages auto exports (State Council, 2009). Total exports were only 0.6 million vehicles in 2012. Since 2008, private firms and local SOEs without JVs have been responsible for almost all passenger vehicle exports, depicted in Appendix Fig. A8.³⁷ Essentially none of these exports were to developed countries. Several high profile Western crash test outcomes help explain the lack of developed country demand. For example, in 2007 Germany and Russia tested Chinese sedans made by Brilliance Jinbei and Chery, respectively. German officials described the Brilliance crash test as “catastrophic,” while the Russian evaluators described the Chery performance as among the worst they had ever encountered (Osborn, 2007).

5.2. Primary regression results

Estimates of Eq. (6) are in Table 2. The standards reduced torque in domestic models relative to foreign models by 11 percent. They reduced price by 15 percent, weight by 65 kg (about 5 percent of the mean), horsepower relative to weight by 6 percent, and power-to-weight by 6.5 percent. All are significant at the 1 percent or 5 percent levels.³⁸ With model fixed effects, the policy's effect on torque declines but remains significant (Panel 1 VI). With firm and class (SUV, minivan, sedan, compact) fixed effects, the results are similar to the main specification (Panel 1 VII–VIII). Controlling for sales volume, in Panel 2 I–IV, has little impact on the coefficient of interest on the interaction between Policy and Domestic. While the policy may affect model sales volume, including it obviates concern that results stem from differential vehicle mix shifting across firm types. The effects become larger and more significant when I expand the bandwidth to all years (1999–2013), in Panel 2 V–IX. This indicates that domestic firms did not catch up to foreign firms in 2012 and 2013.

I split the sample by domestic firm type in Table 3. The policy's effect strengthens when the domestic firm sample is limited to those with JVs (I), at –16 percent for torque, –23 percent for price, and –100 kg for weight. These are significant at the 1 percent level. Among SOEs and SOEs without JVs (II–III), the effects are smaller, but mostly still significantly negative and slightly higher than the full sample effects in Table 2. Among private firms with and without JVs (IV–V), the coefficients are insignificant for all characteristics, and they are near zero for private firms without JVs.

To establish whether these coefficients are significantly different from one another, I estimate Eq. (7) in Table 4. The policy-domestic interaction is modulated with JV and SOE status. The coefficients are relative to foreign firms, and the dummy for being private in the interaction with domestic firm types is omitted. I report the three-way interactions and the quadruple interaction. This estimation, shown in Panel 1, confirms that domestic firms with JVs – rather than simply SOEs – are responsible for the negative effect. The effects are large in magnitude and statistically significant. Panel 2 limits the sample to domestic firms, comparing responses to the policy across domestic firm types. The policy reduced torque by 18 percent, price by 16 percent, and power-to-weight by 8.8 percent among JVs, controlling for the policy's effect on SOEs and the four individual effects. The effect on weight loses significance.

³⁸ Note that the R^2 is very small in most specifications (typically under 10 percent). This is because the Stata procedure (xtreg, fe) treats the groups (e.g. firms) as fixed objects and subtracts them from the model before estimating fit.

³⁴ The data are too noisy at this level of aggregation to include confidence intervals.

³⁵ Similar graphs for weight, not reported for brevity, provide similar visual results.

³⁶ See Melitz and Redding (2012), and Giles and Williams (2000).

³⁷ The biggest exporters are Great Wall (privately-owned, Hebei province-based, listed on the Shanghai stock exchange with no JV), Chery (SOE of the Anhui provincial government with no JV), Geely (privately-owned, listed on the Hong Kong stock exchange with no JV), JAC Motors (SOE majority owned by the Anhui provincial government and partially listed on the Shanghai stock exchange with no JV), and Lifan (privately-owned, listed on the Shanghai stock exchange with no JV). My classification of JV status is by year of sales and ends in 2012. Some companies have since established JVs, such as Chery.

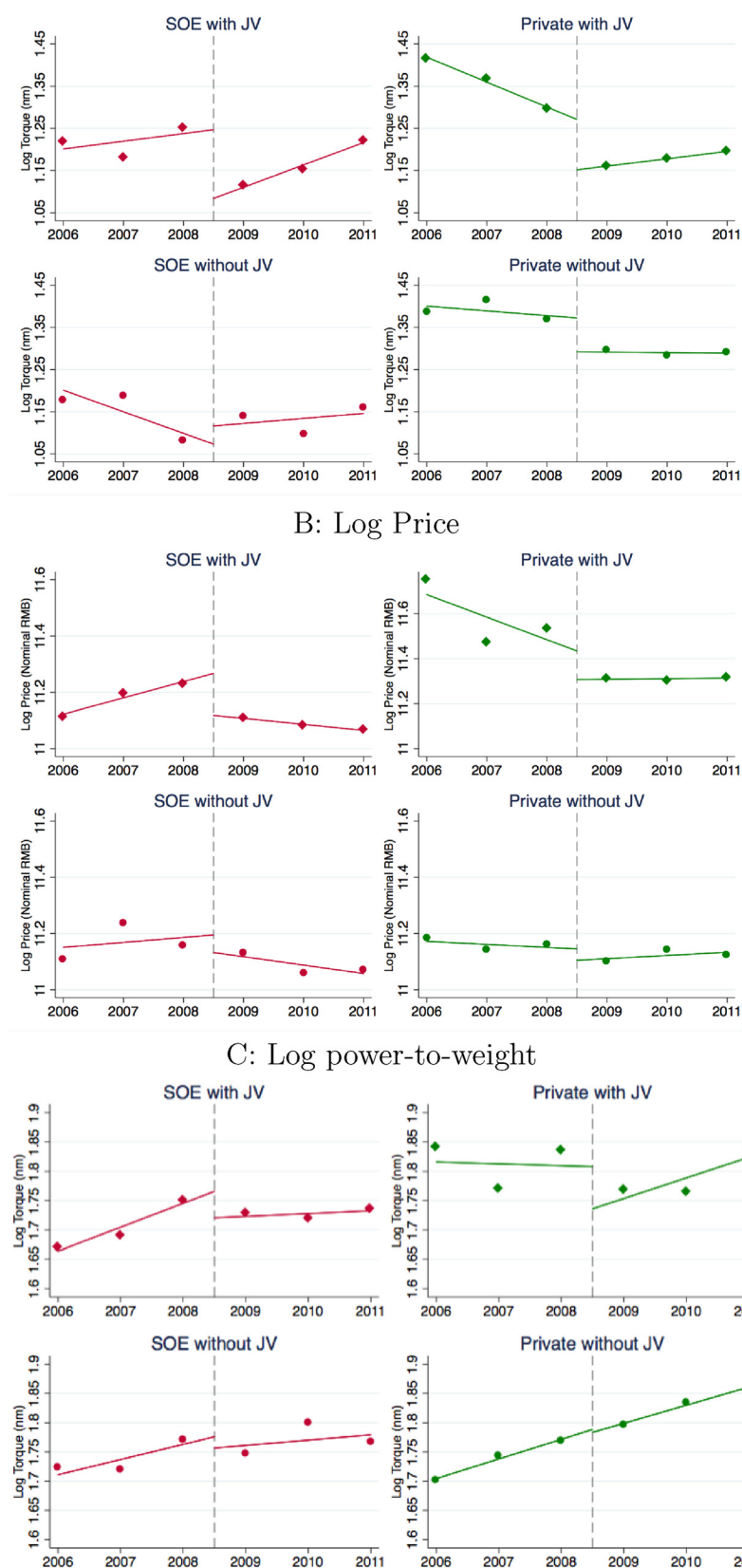


Fig. 5. Domestic firm characteristics by JV and SOE status, 2006–2011. *Note:* This figure shows binned scatterplots by JV status and ownership type.

5.3. Competition results

Next, I show that the policy had stronger effects on firms that competed more intensively with their foreign partners ex-ante. Table 5

I–III limit the sample to the bottom half of the market, considering only models with below-median price, torque, and weight, respectively. The effects increase to –23 percent for torque and price, and –71 kg for weight. IV–VI omit domestic firms with high-end partners. The effects

Table 2
Fuel economy policy impact on domestic firms.

Panel 1: Baseline specification & alternate fixed effects

Dep. variable	Log torque	Log price	Weight	Log HP/weight	Log torque		
	I.	II.	III.	IV.	V.	VI.	VII.
Policy _t ·Domestic _j	−0.11 ^{**} (0.046)	−0.15 ^{***} (0.046)	−65 ^{**} (29)	−0.065 ^{**} (0.028)	−0.065 ^{**} (0.026)	−0.095 ^{**} (0.046)	−0.11 [*] (0.055)
Policy _t	0.037 [*] (0.022)	0.034 (0.022)	31 ^{**} (13)	0.068 ^{***} (0.019)	0.025 (0.018)	0.035 (0.021)	0.04 [*] (0.023)
Domestic _j							−0.18 ^{**} (0.086)
Firm/model/class f.e.	Y/N/N	Y/N/N	Y/N/N	Y/N/N	N/Y/N	Y/N/Y	N/N/Y
N	1643	1653	1599	1595	1643	1643	1643
R ²	0.06	0.2	0.07	0.082	0.900	0.11	0.142

Panel 2: Conditional on sales and all years

Dep. variable	Log torque I.	Log price II.	Weight III.	Log HP/weight IV.	All years (2002–2013)			
					Log torque V.	Log price VI.	Weight VII.	Log HP/weight VIII.
Policy _t ·Domestic _j	−0.1 ^{**} (0.047)	−0.15 ^{***} (0.048)	−60 ^{**} (29)	−0.062 ^{**} (0.029)	−0.13 ^{**} (0.055)	−0.22 ^{***} (0.067)	−90 ^{**} (39)	−0.067 ^{**} (0.031)
Policy _t	0.045 ^{**} (0.022)	0.047 ^{**} (0.022)	37 ^{***} (13)	0.072 ^{***} (0.019)	0.11 ^{***} (0.024)	−0.013 (0.027)	80 ^{***} (17)	0.086 ^{***} (0.014)
SalesVolume _t	−10e−07 ^{***} (3.2e−07)	−2e−06 ^{***} (4.3e−07)	−0.0007 ^{***} (0.00024)	−5e−07 ^{***} (1.9e−07)				
Firm f.e.	Y	Y	Y	Y	Y	Y	Y	Y
N	1643	1653	1599	1595	2032	2062	1986	1974
R ²	0.07	0.13	0.04	0.077	0.076	0.21	0.048	0.018

Note: This table reports difference-in-differences regression estimates of the effect of the 2009 fuel economy standards on domestic firms. Domestic_j is 1 if the brand is domestic (Chinese), and 0 if foreign. Policy_t is 1 if the year is 2009–2–11, and 0 if 2006–2008. In “all years” models must have sales vol of at least 5000. Standard errors are robust and clustered by firm.

** $p < 0.05$.

*** $p < 0.01$.

on all characteristics are again larger than in the main specification. While the magnitudes are larger, they are not always more robust. The effect on power-to-weight, not reported, yielded a large but insignificant effect.

Table 6 contains the class distribution proxy for competition. Panel 1 includes only the subset of JVs where both partners produced above-median levels of SUVs in the three years before the policy. The policy effect is −31 percent for torque, −17 percent for price, and −150 kg for weight. These are all much larger than in the primary full-sample specification in Table 2. Panels 2 and 3 examine partners that specialized in minivans and compacts, respectively, before the policy. They yield somewhat stronger effects on torque than the full-sample specification. While the effects on power-to-weight remain negative, they lose significance. The fuel economy policy was weight-based and thus had different effects on different classes, in particular favoring smaller cars. Such differential overall policy effects should, in the absence of a cannibalization channel, result in the policy having inconsistent effects across classes. That is, the stronger negative effect of the policy on quality among domestic firms producing SUVs should be counteracted by a comparably weaker effect on domestic producers of compact cars. In contrast, I find that the stronger negative effect of the policy on quality persists in all three panels.

In general, the 95 percent confidence intervals for the coefficients in Tables 5 and 6 include the full sample results in Table 4 Panel 1. The precise specifications used do not permit testing whether the coefficients are equal, but in many of the cases I can use slightly different specifications and confidently reject that the coefficients are equal in a χ^2 test.

Thus when partners produced nearer-substitutes pre-policy, the fuel economy standards caused a more extreme reduction in quality and

price. The negative effect of own ϕ_i on the foreign partner's profits ($\frac{\partial \pi_{\text{foreign}, JV}}{\partial \phi_i}$) is stronger when the partner firm produces in similar segments.³⁹ The cannibalization channel appears to outweigh any benefits from knowledge spillovers in the JV. Industrial policy often focuses on increasing or decreasing competition. Perversely, in a JV where the cannibalization channel dominates, more competition with the foreign partner has a negative impact on technology upgrading.

6. Robustness

This section addresses a central concern with my primary empirical design. This is that since JVs are not randomly assigned, a systematic difference across firms unrelated to their JV status could have led to different reactions to the policy. Section 6.1 establishes parallel trends. I provide a more stringent alternative specification of the main result exploiting the staged policy implementation in Section 6.2. Finally, Section 6.3 contains additional robustness tests.

6.1. Pre-policy trends and year-specific effects

It is critical to demonstrate no pre-existing trends at the time of the policy in 2009. If model characteristics across types were on similar growth paths before the policy, the effects that I observe are more readily interpretable as reactions to the policy. Although the Chinese auto industry grew and changed dramatically between 2006 and 2012, the specification is valid if market shocks affected both foreign and

³⁹ I also created a location-based competition index based on the number of foreign JV plants in a given province or city. However, interacting the main effects with this index did not yield systematically different effects across groups.

Table 3
Fuel economy policy impact on domestic firms (firm type splits).

Sample	Firms w/ JVs	SOEs	SOEs w/o JVs	Private	Private w/o JVs
I.	II.	III.	IV.	V.	
<i>Panel 1: Dependent variable = Log torque</i>					
Policy _t ·Domestic _j	−0.16*** (0.061)	−0.12*** (0.047)	−0.13** (0.051)	−0.1 (0.078)	−0.058 (0.093)
Policy _t	0.038* (0.022)	0.037* (0.022)	0.037* (0.022)	0.039* (0.022)	0.039* (0.022)
N	1293	1377	1154	1281	1226
R ²	0.09	0.07	0.04	0.12	0.06
<i>Panel 2: Dependent variable = Log price</i>					
Policy _t ·Domestic _j	−0.23*** (0.066)	−0.21*** (0.057)	−0.19** (0.076)	−0.1 (0.067)	−0.062 (0.074)
Policy _t	0.03 (0.022)	0.031 (0.022)	0.023 (0.022)	0.029 (0.022)	0.029 (0.022)
N	1303	1388	1166	1294	1242
R ²	0.31	0.31	0.26	0.44	0.43
<i>Panel 3: Dependent variable = Weight (kg)</i>					
Policy _t ·Domestic _j	−100*** (26)	−74*** (26)	−61 (49)	−63 (52)	−41 (62)
Policy _t	32** (13)	32** (13)	31** (13)	31** (13)	32** (13)
Firm f.e.	Y	Y	Y	Y	Y
N	1263	1345	1135	1257	1207
R ²	0.07	0.05	0.01	0.1	0.06
<i>Panel 4: Dependent variable = Log HP/weight</i>					
Policy _t ·Domestic _j	−0.097** (0.042)	−0.086*** (0.033)	−0.11*** (0.035)	−0.069* (0.038)	−0.058 (0.041)
Policy _t	0.062*** (0.019)	0.063*** (0.019)	0.06*** (0.019)	0.066*** (0.019)	0.066*** (0.019)
N	1259	1341	1131	1253	1203
R ²	0.079	0.088	0.053	0.048	0.039
Firm f.e.	Y	Y	Y	Y	Y

Note: This table reports difference-in-differences regression estimates of the effect of the 2009 fuel economy standards on domestic firms. Each model includes only domestic firms of the type specified, and all foreign firms. Observations are new model-years. Estimates are variants of Eq. (6); all include firm f.e.. Standard errors are robust and clustered by firm.

*** indicates $p < 0.01$.

domestic firms. I test for statistically different trends over time in model characteristics before the policy in Appendix Table A6. There is no statistically significant difference in trends across firm types (domestic vs. foreign, JV vs. non-JV, and SOE vs. private) prior to the policy. That is, the coefficients on the characteristic interacted with time are all near zero and insignificant. The time trend is mostly positive and slightly significant, reflecting some secular quality improvement before the policy.

In a second specification, I include year fixed effects and interactions around the policy. Eq. (8) estimates the effect of each year, for three years prior to the policy and all years since:

$$Y_{it} = \alpha + \sum_{t=2006}^{2013} (\beta_t \text{Year}_t \cdot X_j + \gamma_t \text{Year}_t) + [1|\text{Firm} = j] + \varepsilon_{ijt}. \quad (8)$$

While there is inadequate power for some post-policy years, the regression results in Appendix Table A6 show (a) parallel pre-trends; and (b) year-specific evidence of the post-policy treatment effect. Although the financial crisis of 2008 did not have a large effect on the Chinese market, the subsequent global recession certainly affected foreign firms in China. The results may reflect foreign pricing pressures as they faced changes in their outside options. Figs. 4 and A5–A7, as well as the regression analysis in this section, indicate that foreign firms continued on their prior path after the policy.

6.2. Triple difference

I confirm the main finding of the negative policy effect on domestic

Table 4
Fuel economy policy impact (firm type interactions).

Dependent variable	Log torque	Log price	Weight	Log HP/weight
I.	II.	III.	IV.	
<i>Panel 1: All firms (effect measured relative to foreign firms)</i>				
Policy _t ·Domestic _j ·Has JV _j	−0.16* (0.083)	−0.2*** (0.052)	−111* (65)	−0.073** (0.029)
Policy _t ·Domestic _j ·SOE _j	−0.045 (0.063)	−0.019 (0.056)	−24 (41)	−0.042 (0.045)
Policy _t ·Domestic _j ·Has JV _j ·SOE _j	0.12 (0.095)	0.12 (0.088)	102 (66)	0.042 (0.044)
Domestic _j ·Has JV _j ·SOE _j	−0.14 (0.23)	−0.26 (0.22)	−259** (124)	0.048 (0.074)
Domestic·Policy & individual effects	Y	Y	Y	Y
Firm & class f.e.	Y	Y	Y	Y
N	1643	1653	1599	1595
R ²	0.34	0.42	0.28	0.19
<i>Panel 2: Domestic firms (effect measured relative to other domestic firms)</i>				
Policy _t ·Has JV _j	−0.18* (0.097)	−0.16*** (0.058)	−114 (85)	−0.088** (0.036)
Policy _t ·SOE _j	−0.11 (0.17)	−0.054 (0.16)	−87 (120)	0.063 (0.045)
Policy _t ·Has JV _j ·SOE _j	0.14 (0.11)	0.071 (0.084)	104 (82)	0.056 (0.063)
Has JV _j ·SOE _j	−0.11 (0.2)	0.095 (0.17)	−111 (109)	−0.046 (0.053)
Individual effects	Y	Y	Y	Y
Firm & class f.e.	Y	Y	Y	Y
N	628	624	596	596
R ²	0.23	0.04	0.22	0.26

Note: This table reports regression estimates of the effect of the 2009 fuel economy standards on domestic firms at the model-year level, using variants of Eq. (7). Panel 2 omits foreign firms and thus all non-identified terms from Eq. (7). Policy_t is 1 if the year is 2009–2011, and 0 if 2006–2008. Standard errors are robust and clustered by firm

** $p < 0.05$.

*** $p < 0.01$.

firms in a triple-differences design that exploits the standards' staged implementation; they applied only to new models in 2008 and all models in 2009. For example, the 2008 Great Wall Peri was a new model as it was not produced in 2007, while the 2008 Volkswagen Jetta was a continuing model. Automakers sensitive to the policy may have changed new model but not continuing model characteristics in 2008. The estimating equation is:

$$Y_{it} = \alpha + \beta (\text{Policy}_t^{2008} \cdot \text{Domestic}_j \cdot \text{Continuing}_{it}) + \gamma_1 (\text{Policy}_t^{2008} \cdot \text{Domestic}_j) + \gamma_2 (\text{Policy}_t^{2008} \cdot \text{Continuing}_{it}) + \gamma_3 (\text{Continuing}_{it} \cdot \text{Domestic}_j) + \gamma_4 \text{Policy}_t + \gamma_6 \text{Continuing}_{it} + (1|\text{Firm} = j) + \varepsilon_{ijt}. \quad (9)$$

The Policy_t²⁰⁰⁸ variable is 1 if the year is 2008, and 0 if 2007 or 2006 (two years are needed for an adequate sample). Here, β is the effect of being a continuing model relative to a new model, netting out the change in means in firm type (domestic vs. foreign) and in time period (after vs. before the 2008 policy). Appendix Table A7 shows that in 2008, domestic firms' continuing models were more powerful, more expensive, heavier, and had higher unobserved quality than new models already subject to the policy, relative to the same comparison within foreign firms. Continuing domestic models not subject to the policy were more powerful, more expensive, and heavier than their new models, relative to the continuing-new difference among foreign firms. Note that the coefficients on the individual indicators and interactions are not direct effects.

Table 5
Fuel economy policy impact on domestic firms with market segmentation (firm type interactions).

Market segment	Models with above-median dependent variable values omitted			Firms with luxury foreign partners omitted ^a		
Dependent variable	Log torque	Log price	Weight	Log torque	Log price	Weight
	I.	II.	III.	IV.	V.	VI.
Policy _{it} · Domestic _j · Has JV _j	−0.23*** (0.047)	−0.23* (0.12)	−71** (35)	−0.22* (0.12)	−0.25*** (0.068)	−196* (102)
Policy _{it} · Domestic _j · SOE _j	0.03 (0.021)	0.033 (0.024)	18* (10)	0.022 (0.021)	0.0045 (0.024)	20 (15)
Policy _{it} · Domestic _j · Has JV _j · SOE _j	0.13*** (0.058)	0.18 (0.13)	17 (28)	0.19 (0.13)	0.17* (0.095)	196* (105)
Domestic _j · Has JV _j · SOE _j	−0.11 (0.078)	−0.11 (0.14)	−2.5 (53)	−0.24 (0.24)	−0.24 (0.24)	−319*** (154)
Domestic-Policy & individual effects	Y	Y	Y	Y	Y	Y
Firm & class f.e.	Y	Y	Y	Y	Y	Y
N	866	850	810	1605	1614	1562
R ²	0.34	0.58	0.22	0.16	0.36	0.25

Note: This table reports difference-in-differences regression estimates of the effect of the 2009 fuel economy standards on domestic firms at the model-year level. Estimates are variants of Eq. (7):

$$Y_{it} = \alpha + \beta_1 (\text{Policy}_{it} \cdot \text{Domestic}_j \cdot \text{Has JV}_{j,t}) + \beta_2 (\text{Policy}_{it} \cdot \text{Domestic}_j \cdot \text{Has JV}_j) + \beta_3 (\text{Policy}_{it} \cdot \text{Domestic}_j \cdot \text{SOE}_j) + \beta_4 (\text{Policy}_{it} \cdot \text{Domestic}_j) + \beta_5 (\text{Domestic}_j \cdot \text{Has JV}_{j,t} \cdot \text{SOE}_j) + \gamma_1 \text{Policy}_{it} + \gamma_2 \text{Has JV}_j + \gamma_3 \text{SOE}_j + [\text{I}] \text{Firm/Class} = j/c + \varepsilon_{ijt}$$

Policy_{it} is 1 if the year is 2009–2011, and 0 if 2006–2008. Standard errors are robust and clustered by firm.

^a Domestic firms that have JVs with BMW or Mercedes-Benz omitted.

* $p < 0.05$.

** $p < 0.05$.

*** $p < 0.01$.

6.3. Further robustness tests

As firms can form new JVs, there may be concern about selection into treatment. Domestic firms may have anticipated the high costs of meeting the standards while maintaining quality, and opted into JVs as a result. This behavior would undermine the empirical design. However, such expectations are at odds with the fact that domestic firms with JVs reduced quality and price in response to the standards. Regardless, Appendix Fig. A4 shows little entry and exit into and out of categories in the three years around the 2009 policy. Only two JVs formed after the standards were announced but before implementation.⁴⁰ There was also no sudden increase in JVs after the standards. Based on Appendix Fig. A4 and anecdotal evidence from the press, JVs were not perceived by domestic firms as a means to deal with the standards, consistent with my conclusion that JVs have not been an efficient technology transfer mechanism.

Next, I conduct key robustness tests of the main result for torque. Appendix Table A8 Panel 1 I–II show the Policy_{it} and Domestic_j interaction and individual effects. The coefficient on the interaction is −7 percent, significant at the 10 percent level, without the individual effects. Panel 1 III omits fixed effects, instead including the Domestic_j individual effect. Panel 1 IV includes both year and firm fixed effects. Year fixed effects should alleviate concerns that the global recession coincided with the policy. China recovered quickly relative to other countries in the second half of 2009, returning to its pre-crisis growth path by 2010 (Diao et al., 2012). Panel 1 V replaces the sales volume requirement of 1000 units with 5000 units. The main finding is intact in all these specifications.

Panel 2 I–III of Appendix Table A8 shows that the result is robust to alternative assumptions about standard errors: brand-year, robust (no clusters), and two-way clustering (Cameron-Gelbach-Miller). Placebo tests in which the policy is artificially set to 2006 and 2005 are in Panel 2 columns IV–V. The coefficient is near-zero for 2006. For 2007 it is −0.077, significant at the 10 percent, which is not surprising as this includes the first part of the policy. In unreported estimation, I do not

find a strong difference between central and local SOEs in their policy response. I do not find effects on length or height. They are less relevant to the quality-fuel economy tradeoff.

7. Conclusion

Understanding the conditions in which firms acquire new technologies is vital to explaining income disparities across countries, and more specifically to evaluating infant industry protection (Parente and Prescott, 1994). I present evidence that the JV mandate in China's auto sector is a distortionary barrier to technology adoption. Conventional trade models like McGrattan and Prescott (2009, 2010) grossly overestimate China's FDI inflows and outflows. They assume that foreign firms bring technological capital to China, which Chinese firms accumulate. When Holmes et al. (2015) add China's requirement that foreign firms transfer technology in order to invest, they are much better able to match their model to moments in the data. They conclude that FDI decreases when foreign firms must transfer technologies. My results confirm this hypothesis: JVs cause foreign firms bring minimum technology to China and disincentivize Chinese firms from investing in technology acquisition.

China's JV mandate, substantial state ownership, and high import tariffs contrast with the Japanese, Taiwanese, and South Korean experiences. Auto sector development in those countries featured an absence of FDI and little direct cooperation with foreign firms, but intensive licensing of foreign technology and reverse engineering.⁴¹ Poorly designed industrial policy may help explain why China's auto sector development has differed so dramatically. A more rigorous WTO regime and tighter IPR protection may also have made it impossible in recent decades to replicate the earlier approaches.

The absence of Chinese exports and the failure of Chinese firms to gain market share suggest that their down-market strategy has not thus far been successful. However, China's automotive industry is changing rapidly. New organizational structures, including independent engineering and design firms that allow domestic automakers to outsource

⁴⁰ These are Chang'an-Mazda and SouEast-Mitsubishi, both announced in 2006.

⁴¹ See Goto and Odagiri (2003).

Table 6
Fuel economy policy impact (pre-policy competition).

Dep. variable	Log torque	Log price	Weight	Log HP/ weight
	I.	II.	III.	IV.
<i>Panel 1: Sample limited to JVs in which both partners produced above-median SUVs before policy</i>				
Policy _{it} Domestic _{it}	−0.31*** (0.1)	−0.17* (0.093)	−149** (71)	−0.055 (0.041)
Policy _{it}	0.12* (0.07)	0.042 (0.066)	32 (39)	0.078*** (0.029)
Domestic _{it}	0.043 (0.14)	−0.7*** (0.11)	−1.9 (91)	−0.25*** (0.065)
N	895	984	876	798
R ²	0.052	0.42	0.043	0.19
<i>Panel 2: Sample limited to JVs in which both partners produced above-median minivans before policy</i>				
Policy _{it} Domestic _{it}	−0.16*** (0.048)	−0.16*** (0.059)	−91*** (35)	−0.057 (0.038)
Policy _{it}	0.051** (0.022)	0.0039 (0.027)	26* (15)	0.081*** (0.023)
Domestic _{it}	−0.2* (0.12)	−0.84*** (0.19)	−123 (105)	−0.24*** (0.06)
N	866	936	846	848
R ²	0.11	0.35	0.034	0.13
<i>Panel 3: Sample limited to JVs in which both partners produced above-median compact cars before policy</i>				
Policy _{it} Domestic _{it}	−0.14*** (0.046)	−0.14** (0.066)	−60* (31)	−0.024 (0.05)
Policy _{it}	0.098*** (0.028)	0.06 (0.037)	49*** (19)	0.086*** (0.021)
Domestic _{it}	−0.18 (0.11)	−0.5*** (0.15)	−28 (74)	−0.13** (0.064)
N	708	746	688	701
R ²	0.15	0.28	0.022	0.11

Note: This table reports estimates of the effect of the 2009 fuel economy standards on domestic firms, using variants of: $Y_{it} = \alpha + \beta(\text{Policy}_{it}\text{Domestic}_{it}) + \gamma_1\text{Policy}_{it} + \gamma_2\text{Domestic}_{it} + \varepsilon_{ijt}$. The sample excludes domestic firms without a JV and further limits the sample to JV partnerships where the domestic and foreign firms both produced above-median levels of a certain class of vehicle prior to the policy (SUVs, minivans, or compacts). Policy_{it} is 1 if the year is 2009–2011, and 0 if 2006–2008. Standard errors are robust and clustered by firm.

* $p < 0.05$.

** $p < 0.05$.

*** $p < 0.01$.

R&D, may enable Chinese firms to undercut foreign competition for small, cheap cars in China and elsewhere. Yet there is evidence that the 2009 situation remains the status quo. According to a Wall Street Journal article, “New proposed [2016] fuel-economy standards for passenger cars...[leave] foreign makers well positioned to inject new technology...That leaves locals such as Great Wall and Geely with the most work to do” (Bhattacharya, 2014).

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.respol.2018.04.021>.

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